



$$\lambda(t) = a + bt \quad \text{avec: } a = 5 \cdot 10^{-4} \quad b = 5 \cdot 10^{-5} \quad T = 100 \text{ (h)}$$

a) SANS MAINTENANCE

$$\begin{aligned} R(T) &= \exp\left(-\int_0^T \lambda(t) dt\right) = \exp\left[-\int_0^T (a + bt) dt\right] = \exp\left[-aT - \frac{bT^2}{2}\right] = \\ &= \exp[-aT] \cdot \exp\left(-\frac{bT^2}{2}\right) = \\ &= \exp(-5 \cdot 10^{-4} \cdot 100) \times \exp\left(-\frac{5 \cdot 10^{-5} \cdot 100^2}{2}\right) = 0,9512 \times 0,7788 = \underline{0,7408} \end{aligned}$$

b) Fiabilité avec 2 maintenance sur $(0, T)$

$$n=2 \quad T_{m+1} = T/3$$

$$\begin{aligned} R_m(T) &= \exp\left[-3 \int_0^{T/3} (a + bt) dt\right] = \\ &= \exp\left[-3 \left(at + \frac{bt^2}{2}\right) \Big|_0^{T/3}\right] = \\ &= \exp\left(-\frac{3aT}{3}\right) \times \exp\left[-\frac{3b}{2} \left(\frac{T}{3}\right)^2\right] = \\ &= \exp(-aT) \times \exp\left(-\frac{bT^2}{6}\right) = \\ &= \exp(-5 \cdot 10^{-4} \cdot 100) \times \exp\left(-\frac{5 \cdot 10^{-5} \cdot 100^2}{6}\right) = \\ &= 0,9512 \times 0,9200 = \underline{0,8751} \end{aligned}$$

c) $n = ?$ pour assurer la fiabilité 0,9 au min

Supposons : $n = 4$ $T/4+1 = T/5$ $n+1 = 5$

$$R_m(T) = \exp \left[-5 \int_0^{T/5} (a + bt) dt \right] =$$
$$= \exp \left[-5 \left(at + \frac{bt^2}{2} \right) \right] \Big|_0^{T/5} =$$

$$= \exp \left[-\frac{5a \cdot t}{5} \right] \cdot \exp \left(-\frac{5b}{2} \frac{T^2}{25} \right) =$$

$$= \exp(-aT) \times \exp \left(-\frac{bT^2}{10} \right) =$$

$$= \exp(-5 \cdot 10^{-4} \cdot 100) \times \exp \left(-\frac{5 \cdot 10^{-5} \cdot 100^2}{10} \right) =$$

$$= 0,9512 \times 0,9512 = \underline{0,9045} > 0,9$$